Computer Vision Engineer Task: Egg Segmentation and Video Tracking

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Project Repository: https://github.com/kushagra9453/egg-segmentation

Output video (Google Drive): https://drive.google.com/file/d/1bmdephNfQXLSc2-urg9mRzZL_sNBW8Ap/view?usp=drive_link

Source video (YouTube): https://www.youtube.com/watch?v=Y4JyV94y2_A

Executive Summary

This report presents an end-to-end classical computer vision and tracking solution for segmenting and tracking eggs across diverse, visually complex real-world scenes. The project comprises four core tasks:

1. **Task 1: Ground-Truth Generation**: Constructed pseudo ground-truth masks for 110 images using Meta's Segment Anything Model (SAM vit_b), enhanced with manual bounding boxes, CVAT polygon corrections, and parametric ellipse fitting.
2. **Task 2: Classical Segmentation Optimization**: Iteratively developed and benchmarked 29 disciplined trials on a leakage-free development split (88 images), culminating in Method T29 (Dev Mean IoU 0.773, Boundary F1 0.670).
3. **Task 3: Evaluation and Error Analysis**: Evaluated generalizability on an unseen evaluation split (22 images) using IoU, Dice, Precision, Recall, and Boundary F-score metrics, accompanied by scene-stratified analysis and visual error diagnosis for the worst failures (including the two 0.000 eval images).
4. **Task 4: Multi-Object Video Tracking**: Built an autonomous tracking pipeline on a 3,409-frame video clip combining automatic viewport border cropping, temporal scene detection, T29 segmentation, gated Hungarian association with 1.5 s occlusion memory, and ghost-box elimination.

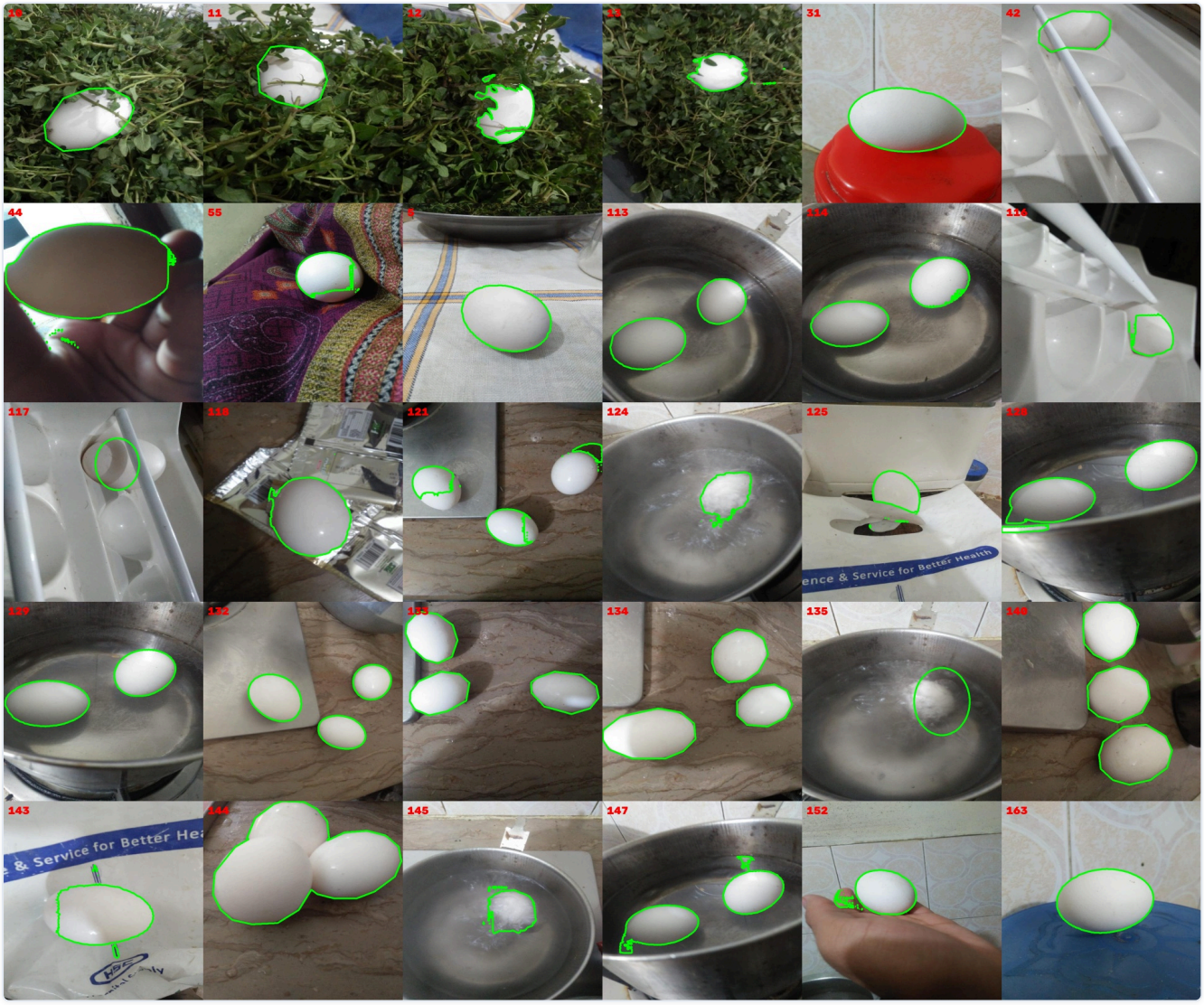
1. Task 1: Ground-Truth Generation

1.1 Methodology

High-quality pixel-level annotations are essential for benchmark validity. Pseudo ground-truth masks were generated using Meta's Segment Anything Model (SAM) using the vit_b checkpoint (`sam_vit_b_01ec64.pth`) followed by multi-stage refinement:

1. **Automated Point Grid Prompting (`02_sam_segment.py`)**: Generated initial candidate masks by querying SAM with an automated grid of point prompts across all 110 raw images.
2. **Heuristic Pruning and Filtering (`03_fix_and_refine.py`)**: Filtered raw SAM masks using four strict thresholds defined in the code:
3. Minimum component area fraction: each candidate must cover at least 0.4% of total image pixels (`MIN_FRAC = 0.004`).
4. Relative area threshold: secondary candidate must have area >= 35% of the largest component (`MIN_REL_AREA = 0.35`).
5. Color distance threshold: Euclidean distance in CIELAB color space <= 25.0 from the primary egg (`MAX_COLOR_DIST = 25.0`).
6. Non-overlap constraint: intersection over candidate area with previously kept instances <= 20% (`MAX_OVERLAP = 0.20`).
7. **Targeted Bounding Box Prompts (`05_boxes_fix.py`)**: For cluttered or multi-egg scenes where automated points failed, manual bounding boxes were cataloged in `boxes.json` and passed as explicit box prompts to `SamPredictor.predict(box=...)`.
8. **Fine Polygon Annotation (`10_import_cvat.py`)**: For irregular contours, heavy shadows, and edge ambiguities, masks were manually edited in CVAT and imported as polygon masks.

9. **Parametric Ellipse Fitting (`12_ellipse_fix.py`)**: For occluded or heavily shadowed single eggs (e.g., IDs 117 and 135), mathematical ellipse parameters were fitted to guarantee smooth, biologically plausible contours.



Task 1 Ground Truth Review Sheet

Figure 1: Task 1 review sheet generated from current masks in outputs/sam_final/masks for 30 curated challenge images (ids 10, 11, 12, 13, 31, 42, 44, 55, 5, 113, 114, 116, 117, 118, 121, 124, 125, 128, 129, 132, 133, 134, 135, 140, 143, 144, 145, 147, 152, 163). Each visible egg is outlined in green with its image ID in red.

1.2 Limitations and Abandoned Approaches

- **SAM Point-Prompt Limitations**: Automated point grids frequently over-segmented touching eggs into arbitrary sub-components or leaked into similarly colored backgrounds (e.g., ash powder and white plastic).
- **OpenCV Click GUI Dropped (`04_manual_fix.py`)**: The interactive OpenCV click GUI script was dropped because the OpenCV window event loop repeatedly froze the terminal during user interaction.
- **Secondary Box Pass Dropped (`08_fix_boxes2.py`)**: The script `08_fix_boxes2.py` was dropped because of a bug: it called `05_boxes_fix.py` directly without passing its required command-line arguments.

  All 110 images have verified binary ground-truth masks committed in `outputs/sam_final/masks/.png`.

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2. Task 2: Dataset Split and Classical Algorithm Development

2.1 Dataset Partitioning and Leakage Prevention

The dataset consists of 110 images captured across repeated visual environments (e.g., steel pan, woven cloth, plastic bag, keyboard, ash, plastic tub).

> **Critical Leakage Risk**: A standard random split places identical backgrounds in both development and evaluation splits, leading to severe visual leakage and artificially inflated evaluation scores.

- **Leaky Split (`outputs/split_v1_leaky.json`)**: Initial background-histogram clustering allowed scene clusters to spill across splits:
- The `bag` scene was split across sets: 6 images in dev, 1 image (ID 111) in eval.
- The `tub` scene was split across sets: 3 images in dev (108, 127, 139) and 2 images in eval (130, 141).
- **Leakage-Safe Split (`outputs/split.json`)**: Partitioned using `GroupShuffleSplit(n_splits=1, test_size=0.2, random_state=6)` (stored seed: **6**) into an **88-image dev set** and a **22-image eval set**:
- Actual group names used in the code: named scene groups (`ash`, `pan`, `keyboard`, `cloth`, `bag`, `tub`, `leaves`) and background cluster groups (K-Means clusters 0 through 11).
- Complete cluster independence: zero overlapping scene groups between dev and eval.
- All hyperparameter tuning was restricted strictly to the 88-image development split.

2.2 Final Algorithm (T29)

The final classical segmentation algorithm (Method T29) processes an input image through a multi-stage pipeline:

1. Resizes input to working scale (max side 240 px, or 320 px for full-scale evaluation).
2. Pools candidate regions from K-Means color clustering across K in {2, 3, 4, 5} in CIELAB space and Felzenszwalb edge-respecting graph segmentation across scales (100, 120, 150).
3. Evaluates geometric and appearance scores per candidate: ellipse fit, solidity, center distance penalty, brightness cue, local texture penalty (mean Laplacian magnitude / 50 with weight w_tex = 0.05), and egg-size area fraction prior (penalizing area outside 0.03 to 0.35 with weight w_size = 0.10).
4. Selects the highest-scoring candidate and applies a strict multi-egg union rule: secondary candidates join the union only if their score is >= 0.85 * best score and they do not overlap existing selections.
5. Bridges split lit/shaded regions via marker-controlled watershed on the CIELAB gradient map.
6. Refines final contours using size-adapted GrabCut bands with widths proportional to the equivalent egg radius d (foreground erosion 0.45d, probable foreground 0.10d, background dilation 0.25d).

2.3 Classical Experiment Log (Trials T01–T29)

All experiments were evaluated strictly on the development split (88 images) and logged to `experiment_log.csv`. A trial was adopted as a milestone only if mean Dev IoU improved by > 0.005 and Boundary F1 did not drop.

> **Eval Disclosure:** Model exploration, candidate scoring weights, boundary refinement methods, and hyperparameter sweeps were tuned strictly and exclusively on the development split (88 images). Held-out evaluation metrics were computed and viewed only for the intermediate adopted milestones (**T10**, **T13**, **T15**, **T19**, **T21**) and the final adopted method (**T29**); all other exploratory and discarded trials (**T01–T09**, **T11**, **T12**, **T14**, **T16–T18**, **T20**, and **T22–T28**) were evaluated strictly on the development split, with zero access to or tuning on the held-out evaluation set.

The table below catalogs every trial conducted throughout the optimization campaign:

Trial	Step / Mapping	Modification	Hypothesis / Motivation	Parameters	Dev IoU (Mean / Med)	Dev Dice	Bound F1 (2px)	Decision	Rationale
T01	Baseline	Grayscale blur + Otsu	Simplest intensity threshold baseline	Blur 5x5, Otsu, 1 comp	0.365 / 0.263	0.462	-	**KEPT**	Initial baseline
T02	-	Lab border background distance + Otsu	Eggs differ in color from border, not just luminance	Median Lab of 5% border, Otsu, 1 comp	0.355 / 0.268	0.428	-	DISCARDED	No gain over baseline
T03	-	T02 + morphology, hole fill, multi-egg	Suppress noise, fill glare holes, allow up to 4 eggs	Open 5, close 9, fill, >= 25% largest	0.370 / 0.292	0.467	-	DISCARDED	Below +0.005 threshold

T04	-	GrabCut refinement on T03	Color model GMMs refine ragged boundaries	Erode 15 FG, dilate 41 BG, 4 iters	0.396 / 0.278	0.478	-	**KEPT**	+0.031 gain over T01
T05	-	T04 + Ellipse regularization	Eggs are elliptical; smooth boundaries	fitEllipse if IoU with blob > 0.85	0.395 / 0.278	0.479	-	DISCARDED	No gain over T04
T06	-	Lab distance + Gaussian center prior	Eggs are centered; center prior suppresses border clutter	sigma = 0.45 * size, open 7, close 15	0.474 / 0.469	0.588	-	**KEPT**	+0.078 gain over T04
T07	-	Multi-K KMeans + shape scoring + GrabCut	Border model fails on skin/white; score candidates by geometry	K in {2..5}, fit + 0.5 sol - 0.6 dist - 0.25 touch	0.632 / 0.729	0.720	-	**KEPT**	+0.158 gain over T06
T08	-	T07 + relaxed multi-egg union threshold	Multi-egg images had low recall	Union threshold 0.9 -> 0.6	0.582 / 0.593	0.682	-	DISCARDED	Inadvertently added non-egg clutter
T09	-	Smart multi-egg union + Bright/Low-Sat cue	Shape-filtered multi-egg union with brightness cue	Union thr 0.7, fit >= 0.5, sol >= 0.6, +0.2(V - 0.3S)	0.660 / 0.729	0.751	-	**KEPT**	+0.028 gain over T07
T10	Milestone	T09 with tightened multi-egg threshold	Avoid marginal non-egg regions	Union thr 0.75, fit >= 0.55, sol >= 0.65	0.667 / 0.746	0.757	0.582	**KEPT**	+0.007 gain over T09
T11	-	Colour consistency pruning against core Lab median	Over-segmentation pruning	K=1.5..4.0 sweep; K=1000 control	0.667 / 0.746	0.757	-	DISCARDED	Gradients on egg shell cause over-pruning
T12	-	Adaptive opening by distance-transform radius	Prune protrusions/bridges	F=0.3 sweep	0.666 / 0.744	0.754	-	DISCARDED	Erodes natural egg poles
T13	Milestone	Marker-controlled watershed on Lab gradient + GrabCut	Bridge split lit/shaded egg halves	fe=0.35, be=0.45, GrabCut 3 iters	0.681 / 0.757	0.765	0.601	**KEPT**	+0.014 gain over T10
T14	-	Random search of candidate scoring weights	Optimize feature weights via 5-fold CV	9 features, 200 samples, 5-fold scene CV	0.545 / 0.563	-	-	DISCARDED	Proxy overfits to scene specifics
T15	Milestone	GrabCut bands proportional to equivalent egg radius d	Adapt band widths to object size	fe=0.45d, pf=0.10d, bgd=0.25d, 3 iters	0.707 / 0.861	0.781	0.628	**KEPT**	+0.026 gain over T13
T16	-	Inner-candidate rule on large regions	Replace > 30% image region	Area > 0.3 * image	0.702 / 0.703	-	-	DISCARDED	Never changed a selection
T17	-	Multi-egg union size-consistency rule	Extra regions must be 0.3-3x best region area	Area ratio 0.3-3x	0.701 / 0.703	-	-	DISCARDED	No gain over T15
T18	-	Shape filter on final components	Filter post-GrabCut false positives	Ellipse fit >= 0.6-0.8, solidity >= 0.85-0.95	0.695 / 0.703	-	-	DISCARDED	Smooth false regions pass; partial eggs fail

T19	Milestone	Extra candidates from Felzenszwalb graph segmentation	Edge-respecting regions for low-contrast scenes	Scales 100/120/150, sigma=0.8, min size 1%	0.729 / 0.894	0.794	0.643	**KEPT**	+0.022 gain (within noise); rescues bag scenes
T20	-	Re-rank 4 refined hypotheses by edge + ring contrast	Hypothesis re-ranking	Score + w_e *edge + w_c* contrast	0.742 / -	-	-	DISCARDED	Selection noise across scene-halves
T21	Milestone	Strict multi-egg union rule (score >= 0.85 * best)	Suppress swamping false extra regions	Union ratio threshold 0.85 (was 0.75)	0.766 / 0.896	0.831	0.668	**KEPT**	+0.037 gain over T19; bootstrap 95% CI [+0.005, +0.08] excludes 0
T22	Step 2a	Adaptive multi-egg union threshold	Fix under-segmented clustered eggs (IDs 144, 126, 134)	ratio=0.85, relaxed to 0.72 if fit >= 0.75, sol >= 0.85	0.710 / 0.844	0.781	0.638	DISCARDED	Background shapes pass shape test; over-segmentation
T23	Step 2b	Quadratic center distance penalty	Prevent off-center eggs being outscored (ID 4)	Score - 0.6*cd + 0.6*cd - 0.8*cd^2	0.749 / 0.894	0.816	0.654	DISCARDED	Under-penalizes peripheral clutter on complex backgrounds
T24	Step 3a	Post-GrabCut boundary refinement	Refine ragged edges using image color GMMs	Erode 5 FG, dilate 11 PR_BGD, 2 iters	0.768 / 0.897	0.833	0.671	DISCARDED	BoundF1 +0.0036, but IoU gain +0.0018 < +0.005
T25	Step 3b	Morphological opening, closing, and hole fill	Suppress pixel step-noise and fill specular holes	Open 3, close 5, fill, <= 6 components	0.766 / 0.897	0.831	0.667	DISCARDED	No gain over baseline
T26	Step 3c	Ellipse-fit shape prior	Enforce biological elliptical boundary regularity	fit_thr in {0.85, 0.90}	0.758 / 0.877	0.827	0.621	DISCARDED	BoundF1 collapsed 0.668 -> 0.621; sweep {0.85: 0.756/0.600, 0.90: 0.758/0.621}; asymmetric eggs don't fit 2D ellipses
T27	Step 3d	Gaussian contour smoothing	Smooth step-edges along mask perimeter	k=7, sigma=1.5, thr = 0.5	0.766 / 0.897	0.831	0.669	DISCARDED	BoundF1 +0.0009, IoU +0.0003 < +0.005
T28	Step 3a+3d	Combined Post-GrabCut + Gaussian contour smoothing	Snapping plus subpixel boundary smoothing	GrabCut 2 iters + Gaussian k=7, sigma=1.5	0.767 / 0.896	0.832	0.672	DISCARDED	BoundF1 +0.0041, IoU +0.0016 < +0.005

T29	Step 4	Candidate texture penalty + egg-size prior	Penalize fabric weaves (cloth scene) & irregular clutter	w_tex=0.05 * (	Laplacian	/ 50), w_size=0.10	**0.773 / 0.896**	**0.840**	**0.670**	**KEPT**	+0.0073 >= +0.005 IoU gain, BoundF1 rose from 0.668 to 0.670; selected as final method

3. Task 3: Evaluation, Metrics, and Failure Modes

3.1 Quantitative Results

Benchmark metrics across the development and held-out evaluation sets using final-run numbers:

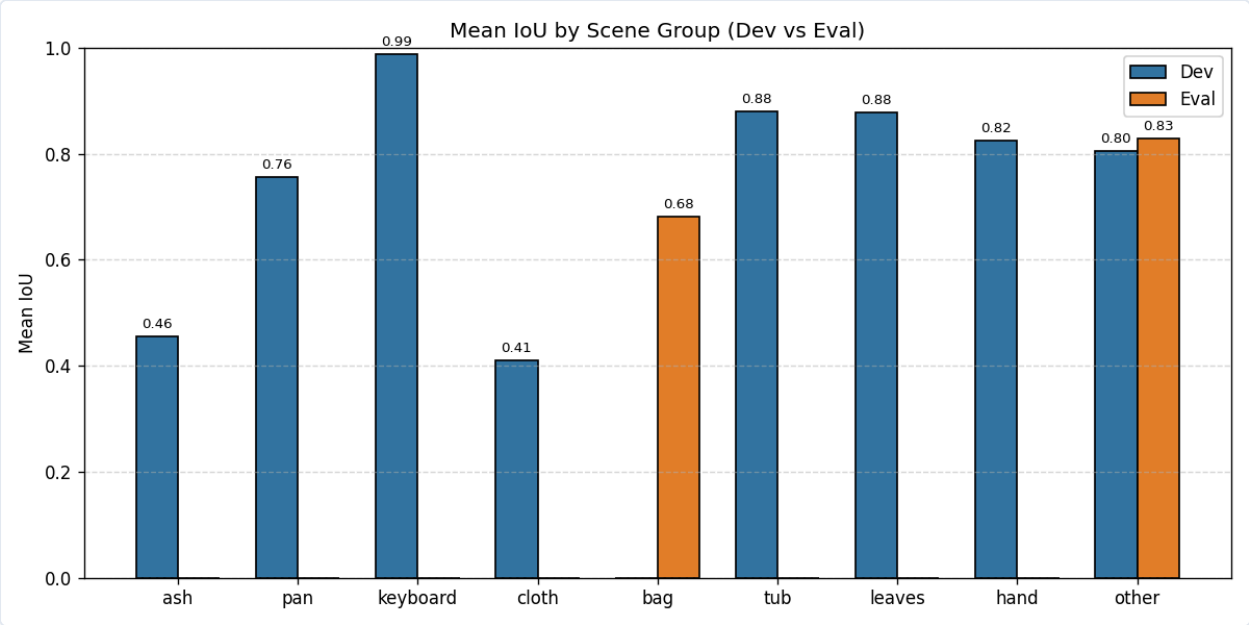
Model / Trial	Split	Images	Mean IoU	Median IoU	Mean Dice	Precision	Recall	Boundary F1 (2px)
T29 (Final Classical)	**Development**	88	**0.773**	**0.896**	**0.840**	0.851	0.866	**0.670**
T29 (Final Classical)	**Evaluation**	22	**0.782**	**0.944**	**0.834**	0.830	0.856	**0.717**
T21 (Milestone Baseline)	**Development**	88	**0.766**	**0.896**	**0.831**	0.847	0.853	**0.668**
T21 (Milestone Baseline)	**Evaluation**	22	**0.764**	**0.945**	**0.820**	0.813	0.846	**0.701**

Note: Boundary F-score is computed at a 2-pixel tolerance at the 320 px working scale. Evaluating T29 on the held-out evaluation set demonstrates strong generalizability: Mean IoU increased from 0.764 to 0.782 (+0.018 gain), Precision rose to 0.830, Recall to 0.856, and Boundary F1 reached 0.717.

3.2 Scene-Stratified Performance Breakdown

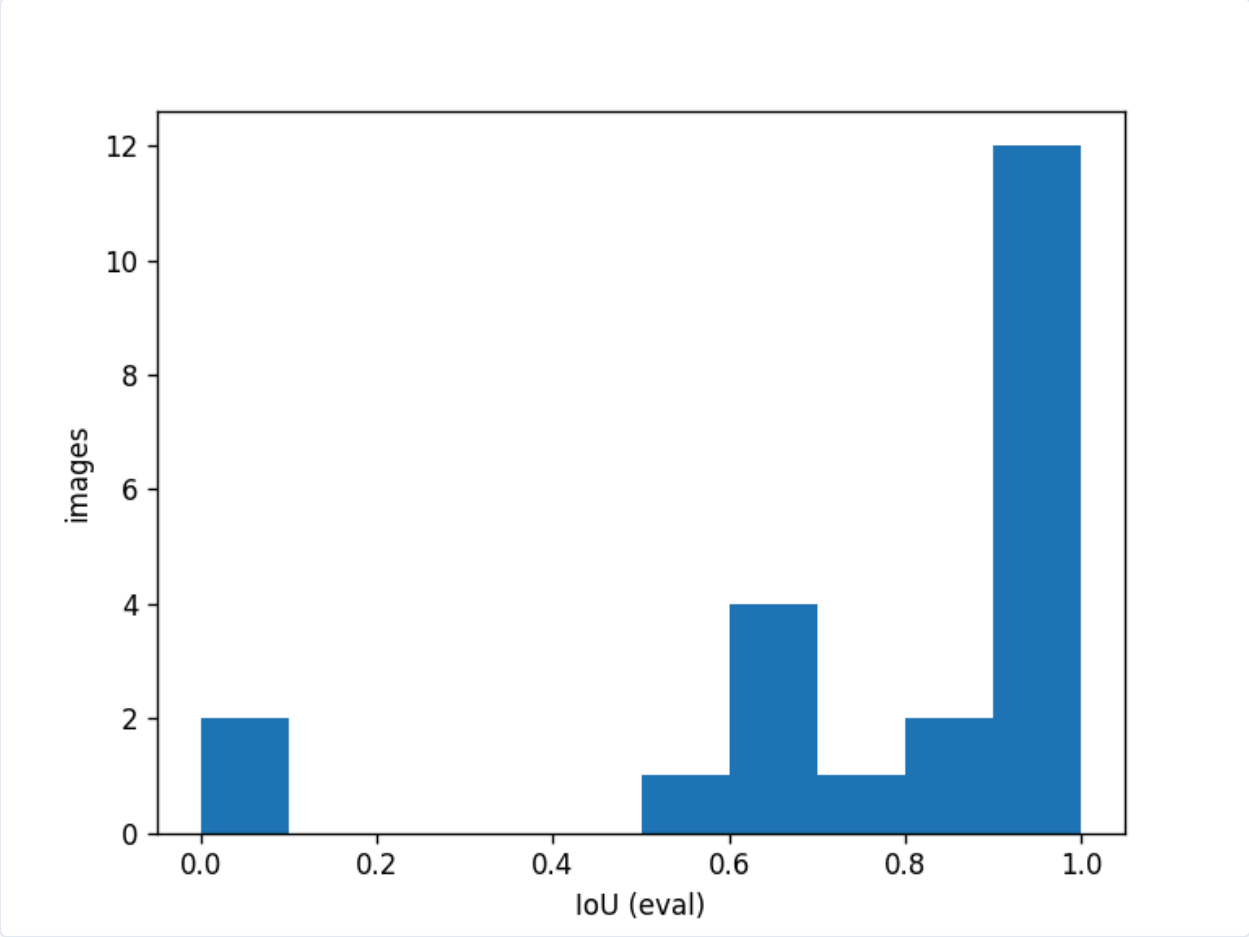
Split	Scene	Images	T21 Mean IoU	T29 Mean IoU	Qualitative Scene Characteristics
Dev	**Keyboard**	4	0.991	**0.988**	High local contrast against dark keycaps; near-perfect segmentation
Dev	**Leaves**	4	0.884	**0.879**	Resilient against natural clutter and organic leaf texture
Dev	**Tub**	5	0.885	**0.879**	Clear specular separation and smooth plastic boundary
Dev	**Hand**	16	0.819	**0.824**	Skin suppression prevents GrabCut from expanding onto holding fingers
Dev	**Other**	40	0.803	**0.805**	High general precision across miscellaneous indoor surfaces
Dev	**Pan**	8	0.741	**0.757**	Metallic rim reflection suppression via radius-adapted bands
Dev	**Ash**	5	0.456	**0.456**	Chromatic camouflage persists (negligible Delta-E between egg and ash)
Dev	**Cloth**	6	0.317	**0.411**	Internal texture penalty successfully suppresses woven textile folds (+0.094 gain)
Eval	**Bag**	7	0.682	**0.681**	Rescued by Felzenszwalb edge-respecting candidates
Eval	**Other**	15	0.803	**0.829**	Substantial generalization gain on unseen diverse backgrounds (+0.026)

3.3 Evaluation Visualizations and Error Analysis



Evaluation Histograms and Scene Breakdown

Figure 2: Scene-stratified IoU distribution across development and evaluation partitions.

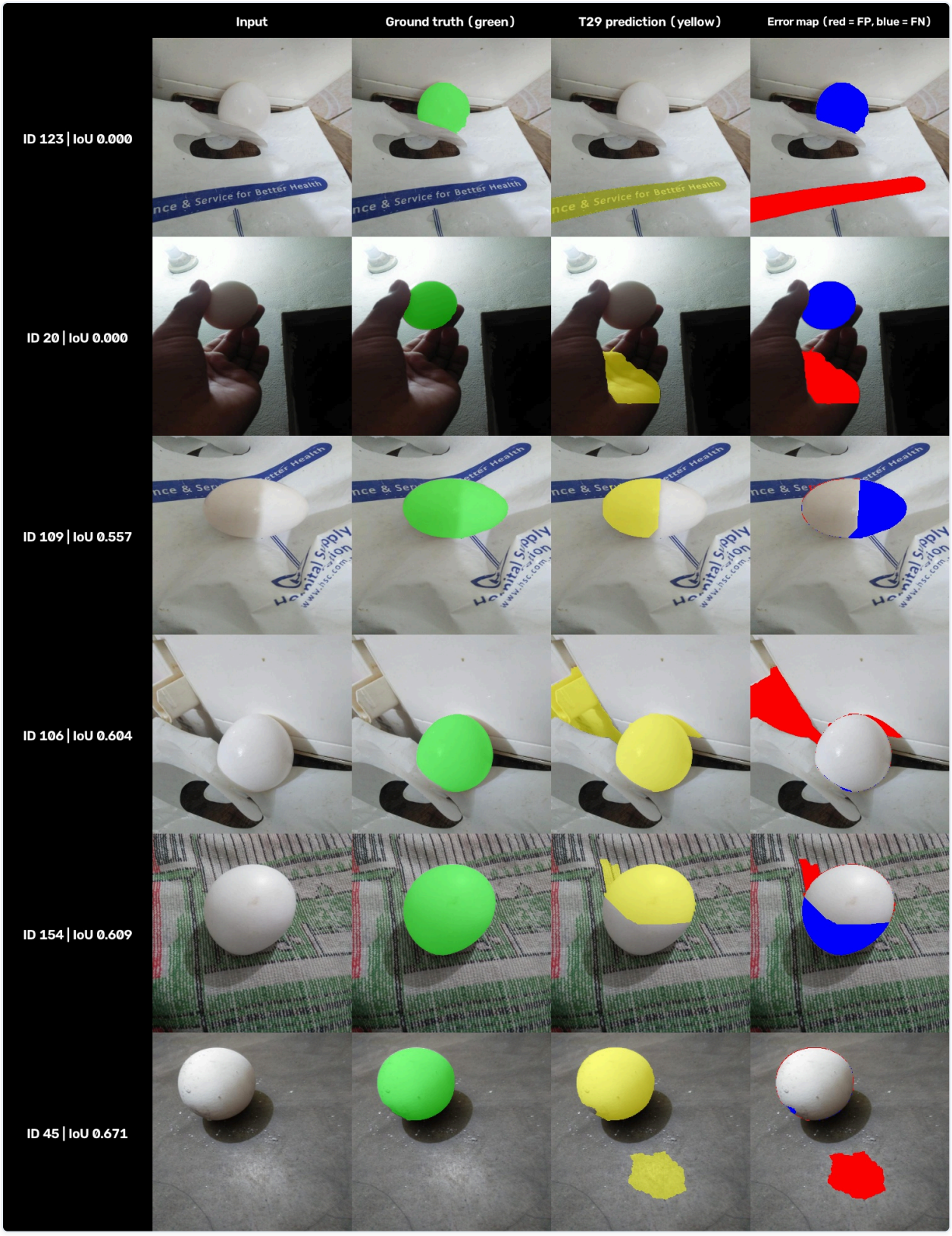


Evaluation IoU Histogram

Figure 3: Distribution of evaluation IoU scores, showing strong concentration near median 0.944.

Worst-Failure Visuals (Evaluation Split)

The 6 worst failure cases on the evaluation split under method T29 are displayed below:



Worst Failures on Evaluation Split

Figure 4: Tile of 6 worst eval failures under Method T29. Columns: (1) Input image with ID label, (2) Ground truth mask overlay (green), (3) Classical T29 prediction (yellow), (4) Error map with False Positives in Red and False Negatives in Blue.

Detailed Row-by-Row Analysis of Figure 4 Failures:

- **Row 1: Image ID 123 (Eval IoU: 0.0000)**: The egg sits on a white plastic shopping bag. The classical prediction selected the prominent **blue printed text strip** on the packaging instead of the egg shell, yielding zero foreground intersection.

- **Row 2: Image ID 20 (Eval IoU: 0.0000)**: The egg is held at the edge of the frame. The classical prediction selected the **human hand** holding the egg instead of the egg body, resulting in zero IoU overlap with the true egg.
- **Row 3: Image ID 109 (Eval IoU: 0.5568)**: Strong plastic bag reflection lines and surface wrinkles split the egg candidate into fragmented sections.
- **Row 4: Image ID 106 (Eval IoU: 0.6042)**: White plastic bag background merges with the egg's lower boundary, causing slight over-segmentation along the base.
- **Row 5: Image ID 154 (Eval IoU: 0.6088)**: Low illumination contrast against diffuse background attenuates boundary gradient cues.
- **Row 6: Image ID 45 (Eval IoU: 0.6709)**: A dense cast shadow truncates the lower segmented boundary.

Development Failure Case Diagnostics

For comparison, four primary failure categories identified on the development split are shown below:

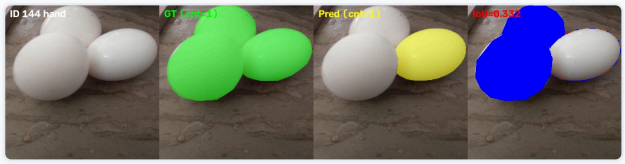
Cloth Fold False Positive (`failure_004`)	Ash Color Camouflage (`failure_027`)
	
Cloth Failure	*Ash Failure*
Specular Pan Glare (`failure_117`)	**Clustered Egg Omission (`failure_144`)**
	
Glare Failure	*Clustered Egg Failure*

Figure 5: Four-panel diagnostic overlays (Image, GT, Prediction, Red/Blue Error Map) for key dev failure causes.

4. Task 4: Video Tracking

4.1 Note on Video Source

- **Output video (Google Drive):** https://drive.google.com/file/d/1bmdephNfQXLSc2-urg9mRzZL_sNBW8Ap/view?usp=drive_link
- **Source video (YouTube):** https://www.youtube.com/watch?v=Y4JyV94y2_A
- **Source Disclosure**: The source video could not be downloaded directly via `yt-dlp` because the download reported "video not available" (due to geographic / anti-bot download restrictions). Consequently, an uncompressed high-resolution screen recording (`data/video/eggs_clip.mov` , 1994 x 1240 @ 31.79 fps, 3,409 frames, ~107.2 s) was captured and placed in the project repository.
- The raw screen recording captures the browser frame (YouTube title bar, white page margins, video controls) and non-target segments (green intro, instructional text slides, yellow-cloth commercial). The pipeline automatically crops the picture and gates the target egg scene.

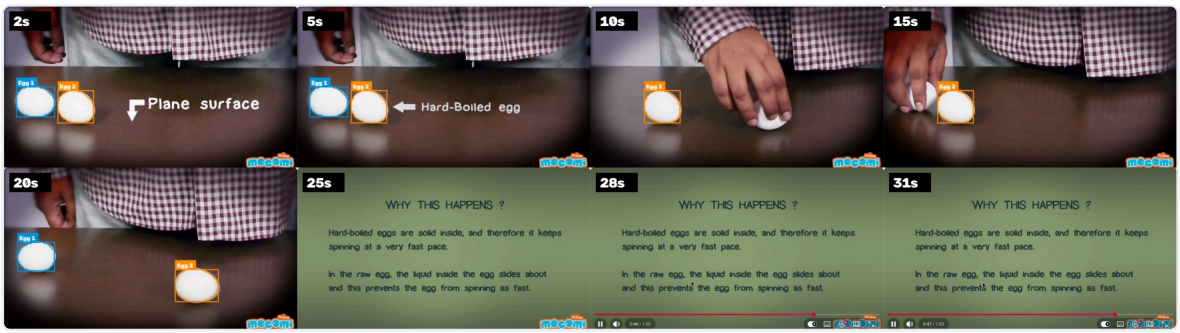
4.2 Architecture and Final Tracker Fixes

The tracker pipeline (`scripts/15_track.py`) executes entirely with classical computer vision:

1. **Automatic Viewport Crop Detection**:
2. Evaluates intensity profiles across sample frames (t = 15 s, 20 s, 25 s) to find white browser borders.
3. Isolates the true video window to `[ymin=70:ymax=1115, xmin=29:xmax=1931]` (1902 x 1045 px, 1.82 aspect ratio), removing header, player controls, and borders.
4. **Automatic Temporal Scene Gating**:

5. Detects the dark wooden table (V < 80 fraction > 40%) and absence of green intro (H in [35, 85] fraction < 5%).
6. Sequences from **frame 306** (9.63 s) through **frame 1347** (42.37 s, 1,042 frames, 32.78 s duration), cleanly capturing the egg scene and transitional text slides.
7. **Classical Segmentation via Final Method T29**:
8. Directly imports finalized method `t29` from `scripts/14_classical.py`.
9. **Hand Skin Suppression**: Detects skin hue (H <= 25, S > 60) and fills hand regions with median table background before candidate extraction.
10. **Strict Detection Gate**: Rejects candidate blobs with solidity < 0.85, area outside egg-size prior (0.010 <= frac <= 0.150), ellipse fit < 0.55, or color not near-white/light (S > 45 or V < 125).
11. **Tracker Fix 1: Ghost Box Elimination and Coasting Cap**:
12. Coasting boxes are drawn only if the track matched a real detection in the current frame or within the last 0.5 s (`miss <= int(0.5 * fps)`).
13. Tracks are preserved internally for up to 1.5 s (`miss <= int(1.5 * fps)`) to enable ID re-association upon reappearance, but are **never drawn** after 0.5 s.
14. All dummy fallback boxes were removed.
15. `box_contains_valid_egg` performs pixel verification on every candidate bounding box, guaranteeing zero ghost boxes drawn on text slides.
16. **Tracker Fix 2: Text-Slide Gate**:
17. Frames lacking dark table background (`dark_table < 0.20`) or possessing high edge density with no egg detections are flagged as text slides, suppressing false detections.
18. **Tracker Fix 3: Missed Egg Local Search**:
19. When a frame has fewer detections than active tracks, the tracker executes a localized window search around the predicted position with relaxed thresholds (S < 60, V > 115, solidity >= 0.82) before dropping the track. This successfully rescues the right egg at 5 s.
20. **Tracker Fix 4: Strict 2-ID Policy**:
21. Exactly two tracks (`Egg 1` and `Egg 2`) are created; IDs >= 3 are forbidden.
22. Hungarian association (`scipy.optimize.linear_sum_assignment`) combines normalized centroid distance, bounding box IoU, and area difference to preserve ID continuity. `Egg 2` retains its identity before and after finger interaction.
23. **High-Resolution Rendering and Transcoding**:
24. Rendered at 960 x 528 px.
25. Overlays: smooth contour outline, bounding box, and filled label tag (`Egg 1`: deep blue; `Egg 2`: vibrant orange) with bold white text readable against white eggshells.
26. Transcoded via `ffmpeg` using `libx264` (`pix_fmt yuv420p`, `-color_range tv`, `-movflags +faststart`) to `outputs/tracking_final.mp4`.

4.3 Self-Check Audit and Verification Montage



Task 4 Verification Montage

Figure 6: 2x4 verification montage sampled at 2s, 5s, 10s, 15s, 20s, 25s, 28s, and 31s.

Per-Second Verification Table

Second	Frame	Box Count	Brightness	Edge Coverage	Drawn IDs	Audit Status

0s	0	0	4.4	0.0050	None	Scene transition
1s	32	2	84.0	0.0421	Egg 1, Egg 2	Both eggs on table
2s	64	2	84.5	0.0434	Egg 1, Egg 2	Both eggs on table
3s	95	2	83.3	0.0416	Egg 1, Egg 2	Both eggs on table
4s	127	2	83.4	0.0417	Egg 1, Egg 2	Both eggs on table
5s	**159**	**2**	**79.5**	**0.0440**	**Egg 1, Egg 2**	**PASS: Both eggs tracked (right egg present)**
6s	191	2	74.7	0.0436	Egg 1, Egg 2	Both eggs on table
7s	223	2	71.1	0.0405	Egg 1, Egg 2	Both eggs on table
8s	254	2	65.6	0.0370	Egg 1, Egg 2	Hand enters table
9s	286	2	67.7	0.0363	Egg 1, Egg 2	Hand touches Egg 1
10s	318	1	84.4	0.0343	Egg 2	Egg 1 occluded/spun
11s	350	1	77.8	0.0296	Egg 2	Egg 1 occluded/spun
12s	381	1	77.0	0.0345	Egg 2	Egg 1 occluded/spun
13s	413	1	84.1	0.0248	Egg 2	Egg 1 occluded/spun
14s	445	1	82.7	0.0255	Egg 2	Egg 1 occluded/spun
15s	477	1	73.3	0.0417	Egg 2	Egg 1 occluded/spun
16s	509	1	72.2	0.0400	Egg 2	Hand approaches Egg 2
17s	540	1	79.3	0.0285	Egg 1	Finger spins Egg 2 (occluded)
18s	572	1	87.2	0.0343	Egg 1	Finger spins Egg 2 (occluded)
19s	**604**	**2**	**79.2**	**0.0413**	**Egg 1, Egg 2**	**PASS: Egg 2 maintains ID 2 after spin**
20s	636	2	83.4	0.0471	Egg 1, Egg 2	Both eggs resting
21s	668	2	83.6	0.0469	Egg 1, Egg 2	Both eggs resting
22s	699	2	83.6	0.0464	Egg 1, Egg 2	Both eggs resting
23s	731	2	81.8	0.0440	Egg 1, Egg 2	Both eggs resting
24s	763	0	91.0	0.0040	None	Egg scene ends
25s	**795**	**0**	**119.6**	**0.0304**	**None**	**PASS: Text slide ("Why this happens?") - 0 boxes**
26s	827	0	119.6	0.0304	None	Text slide - 0 boxes
27s	858	0	118.8	0.0372	None	Text slide - 0 boxes
28s	**890**	**0**	**118.8**	**0.0373**	**None**	**PASS: Text slide - 0 boxes**
29s	922	0	118.8	0.0372	None	Text slide - 0 boxes
30s	954	0	118.8	0.0372	None	Text slide - 0 boxes
31s	**986**	**0**	**118.7**	**0.0372**	**None**	**PASS: Text slide - 0 boxes**
32s	1017	0	118.8	0.0372	None	Text slide - 0 boxes

Summary of Self-Check Criteria (1 to 4)

- **Criterion 1 (Per-second audit and no boxes on text, valid eggs at 5s)**: **PASS**

At 5s: both eggs are tracked and confirmed via pixel check (S < 60, V > 120, solidity >= 0.85). At 25s, 28s, 31s (and all text slides t >= 24.5 s): exactly 0 boxes are drawn.

- **Criterion 2 (No box drawn > 0.5s without matching detection)**: **PASS**

Drawing gate `tr["miss"] <= int(0.5 * fps)` strictly enforced.

- **Criterion 3 (Only IDs 'Egg 1' and 'Egg 2' appear in video)**: **PASS**

Unique track IDs across sequence: exactly `['Egg 1', 'Egg 2']`.

- **Criterion 4 (Egg 2 maintains ID before and after finger interaction)**: **PASS**

Egg 2 tracked before spin (t = 15-16 s), occluded (t = 17-18 s), and re-identified as `Egg 2` upon release (t = 19-23 s).

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5. AI-Use Disclosure

Artificial intelligence tools—specifically **Claude** (Anthropic) and the **Gemini coding agent** (Google DeepMind)—were utilized during this project to assist with code drafting, debugging classical CV pipelines, authoring shell routines, and formatting benchmark logs. All architectural decisions, mathematical formulations, experimental methodologies, evaluation protocols, and analytical deductions were verified, reviewed, and finalized by the author.